```python
import threading
import time
import random
BUFFER_SIZE = 5
buffer = []
empty = threading.Semaphore(BUFFER_SIZE)
full = threading.Semaphore(0)
mutex = threading.Semaphore(1)
def producer():  1 usage
    for i in range(10):
        item = random.randint(1, 100)
        empty.acquire()
        mutex.acquire()
        buffer.append(item)
        print("Produced:", item)
        print("Buffer:", buffer)
        mutex.release()
        full.release()
        time.sleep(1)
```

```python
def consumer():  1 usage
    for i in range(10):
        full.acquire()
        mutex.acquire()
        item = buffer.pop(0)
        print("Consumed:", item)
        print("Buffer:", buffer)
        mutex.release()
        empty.release()
        time.sleep(1)
t1 = threading.Thread(target=producer)
t2 = threading.Thread(target=consumer)
t1.start()
t2.start()
t1.join()
t2.join()
print("Producer-Consumer execution completed")
```

```
Produced: 80
Buffer: [80]
Consumed: 80
Buffer: []
Produced: 3
Buffer: [3]
Consumed: 3
Buffer: []
Produced: 1
Buffer: [1]
Consumed: 1
Buffer: []
Produced: 69
Buffer: [69]
Consumed: 69
Buffer: []
Produced: 48
Buffer: [48]
Consumed: 48
Buffer: []
Produced: 32
Buffer: [32]
Consumed: 32
Buffer: []
Produced: 84
Buffer: [84]
Consumed: 84
Buffer: []
```

```
Consumed: 23
Buffer: []
Produced: 25
Buffer: [25]
Consumed: 25
Buffer: []
Produced: 66
Buffer: [66]
Consumed: 66
Buffer: []
Producer-Consumer execution completed
```